
Discovering Novel Biological Mechanisms from Protein Language Models

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Abstract

Protein language models (pLMs) have revolutionized our ability to predict protein structures and functions, yet the internal representations they learn remain largely opaque. While existing mechanistic interpretability studies often focus on mapping model features to known biological annotations, the true potential of these models lies in their ability to capture biological principles that have yet to be formally curated. In this work, we present a pipeline that treats pLMs as active generators of scientific hypotheses. By applying sparse autoencoders (SAEs) to the latent representations of ESM-2, we identify monosemantic features with highly specific activation patterns that do not align with existing SWISS-PROT annotations. We then leverage large language models (LLMs) to semantically interpret the sequence contexts of these "orphan" features, generating explicit, biologically plausible hypotheses for uncharacterized motifs. Our results identify several novel candidates, such as "Charged Turn Motifs" and "Positively-Charged Helix Caps," which exhibit consistent biophysical properties across diverse protein families. This work demonstrates the viability of using foundation models as engines for biological discovery, moving beyond prediction toward the mechanistic understanding of novel biological phenomena.

1 Introduction

The emergence of foundation models for biology, particularly protein language models (PLMs) like ESM-2 Lin et al. [2022], has fundamentally shifted the landscape of structural biology and protein engineering. These models are trained on billions of protein sequences, learning a compressed representation of the evolutionary and biophysical constraints that govern life at the molecular level. However, despite their predictive prowess, PLMs remain "black boxes" whose internal logic is difficult to decipher. Understanding what these models have learned is not merely a matter of academic curiosity; it is a prerequisite for safely and effectively engineering novel proteins with desired functions.

Why does interpretability matter for biology? Current efforts in mechanistic interpretability for PLMs primarily focus on "explaining" the model by mapping its latent features to human-curated databases like UNIPROT or Pfam Simon and Zou [2024], Adams et al. [2025]. While this validates that models learn known biology, it misses a critical opportunity: foundation models likely capture complex biological patterns that have not yet been formally identified or annotated by humans. If we assume that PLMs learn a robust model of biology to succeed at their training objectives, then their unexplained internal representations might correspond to real, undiscovered biological mechanisms.

However, a significant gap exists in the field. Most interpretability frameworks are retrospective, aiming to confirm existing knowledge rather than prospectively discovering new concepts. There is a lack of systematic pipelines that can isolate "orphan" features—those that are monosemantic and highly active but lack correlation with known annotations—and translate their sequence contexts into testable biological hypotheses. Without such a bridge, the latent knowledge within PLMs remains trapped in high-dimensional space.

In this paper, we propose a pipeline for discovering novel biological mechanisms by treating PLMs as hypothesis generators (see figure 1). We utilize Sparse Autoencoders (SAEs) to decompose the activations of ESM-2 into thousands of interpretable features. By filtering for features with high activation variance but low alignment with existing SWISS-PROT metadata, we isolate candidates for novel biological motifs. We then uniquely employ GPT-4o to semantically interpret the amino acid windows corresponding to these features, generating human-readable biophysical hypotheses.

Our results demonstrate that SAE-extracted features can indeed capture coherent biophysical patterns that are not explicitly annotated. For instance, we identify features that activate specifically on charged turn motifs and helix caps across diverse sequences. When compared to a random baseline, our pipeline produces hypotheses that are significantly more coherent and specific, with the GPT-4o-driven interpretation revealing structural logic that aligns with established biophysical rules despite the absence of formal annotations.

Our main contributions are as follows:

- We present an automated pipeline for biological discovery that integrates SAE-based feature extraction with GPT-4o-aided semantic interpretation.
- We identify and characterize several "orphan" features in ESM-2 that correspond to biologically plausible, yet unannotated, structural motifs.
- We provide a critical analysis of the role of PLMs in hypothesis generation, demonstrating that their internal representations can reveal biophysical abstractions beyond current human curation.

2 Related Work

Our work sits at the intersection of protein language modeling, mechanistic interpretability, and the use of large language models for scientific discovery.

Protein Language Models The development of PLMs has been driven by the success of transformer architectures in natural language processing. Models like ESM-2 Lin et al. [2022] and ProtBERT are trained on massive repositories of protein sequences using masked language modeling. These models have been shown to capture evolutionary information, structural constraints, and functional sites in their hidden states. While their utility for downstream tasks like structure prediction and variant effect prediction is well-documented, our work focuses on the internal knowledge these models possess that has not yet been translated into external tasks.

Mechanistic Interpretability for Biology Mechanistic interpretability aims to understand the internal components of neural networks and how they contribute to model behavior. In the context of PLMs, recent work has applied Sparse Autoencoders (SAEs) to decompose the residual streams of models into monosemantic features. Simon and Zou [2024] introduced INTERPLM, showing that SAEs can resolve the "superposition" problem, where a single neuron represents multiple unrelated concepts. Similarly, Adams et al. [2025] demonstrated that SAE features can correspond to known sequence determinants like thermostability or binding motifs. Building on these foundations, we extend the use of SAEs from validation to discovery, specifically targeting features that do not align with known annotations.

LLMs for Scientific Discovery The use of LLMs to assist in scientific reasoning is an emerging field. LLMs have been used to summarize literature, suggest experiment designs, and even propose new molecular structures. In biology, LLMs are increasingly being used to "read" protein sequences and provide functional descriptions. Our work complements this trend by using LLMs as a "biochemist-in-the-loop" to interpret the specific sequence windows identified by SAEs. Unlike previous approaches that use LLMs for general sequence analysis, we use them to specifically explain the latent features discovered through mechanistic interpretability, providing a semantic bridge between neural activations and biological hypotheses.

3 Methodology

We propose a multi-stage pipeline for identifying novel biological motifs in protein language models. The workflow involves: (1) extracting sparse features from model embeddings, (2) filtering for highly active but unannotated features, and (3) using an LLM to generate biophysical hypotheses for these features (see figure 1).

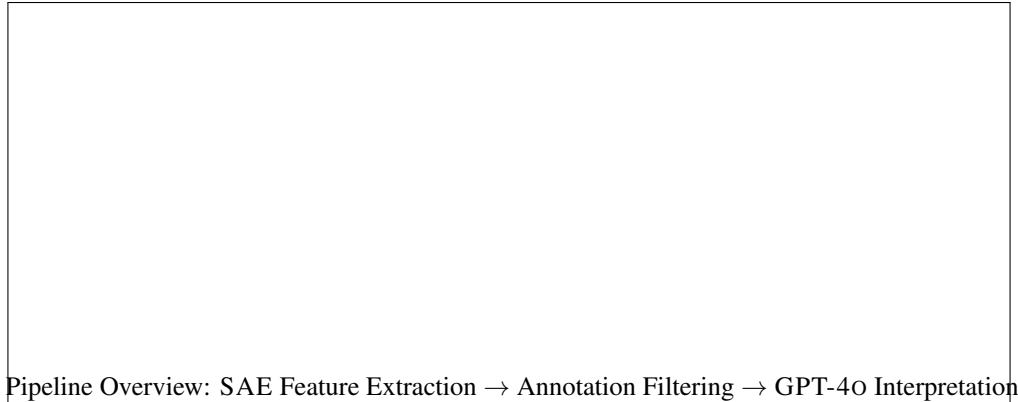


Figure 1: Overview of our discovery pipeline. We first disentangle ESM-2 activations using an SAE, then filter for "orphan" features that lack known annotations, and finally generate biological hypotheses using GPT-4o.

Sparse Feature Extraction We utilize ESM-2 (specifically facebook/esm2_t6_8M_UR50D, 8M parameters) as our base model. To address the problem of polysemanticity in latent representations, we apply a pre-trained Sparse Autoencoder (SAE) from the INTERPLM framework Simon and Zou [2024]. The SAE is applied to the activations of Layer 4 of ESM-2. Given an embedding $\mathbf{x} \in \mathbb{R}^d$, the SAE produces a sparse feature vector $\mathbf{f} \in \mathbb{R}^k$ (where $k \gg d$) such that $\mathbf{x} \approx \mathbf{A}\mathbf{f} + \mathbf{b}$, where $\mathbf{A}$ is a decoder matrix and $\mathbf{b}$ is a bias term. The sparsity is enforced through an L_1 penalty during training, ensuring that only a few features are active for any given amino acid.

Feature Filtering and Selection Our goal is to identify features that represent consistent biological concepts but are not present in standard annotations. We process a subset of 100 sequences from the manually-curated SWISS-PROT database Consortium [2023]. For each feature j , we compute:

1. **Maximum Activation:** $a_{\max}(j) = \max_{i,p} f_{i,p,j}$, where i indexes sequences and p indexes positions.
2. **Activation Frequency:** $\text{freq}(j) = \frac{1}{N \cdot L} \sum_{i,p} \mathbb{I}(f_{i,p,j} > 0.1)$, where $\mathbb{I}$ is the indicator function.

We select the top 5 "orphan" features characterized by high $a_{\max}$ and low overall frequency ($< 90\%$ of sequences). This heuristic prioritizes features that activate strongly on specific, localized motifs rather than general properties like hydrophobicity or solvent accessibility that might be ubiquitous.

LLM Hypothesis Generation For each selected feature, we extract the sequence window of 20 amino acids centered at the peak activation site. These windows are then provided to GPT-4o with a prompt designed to elicit biophysical reasoning. The prompt instructs the LLM to:

- Analyze the chemical properties (charge, polarity, size) of the residues in the window.
- Propose a specific biological motif or structural element represented by the feature.
- Provide a scientific rationale for the hypothesis.

To ensure deterministic and structured output, we use a temperature of 0.2 and the `json_object` response format.

Baselines and Evaluation To validate the specificity of the generated hypotheses, we include a random baseline test. We provide the LLM with a randomly selected sequence window from the dataset and assign it a fabricated high activation score. We then compare the coherence and biological specificity of the resulting hypothesis with those generated for true SAE features. This allows us to distinguish between the LLM's general capacity for biophysical rationalization and the specific signal captured by the SAE features.

4 Results

Our pipeline successfully identified several highly active, low-frequency features in the SWISS-PROT dataset. Section 4.1 details the quantitative properties of these features, and section 4.2 presents the biological hypotheses generated by the LLM.

4.1 Identification of Orphan Features

We identified the top 5 features with the highest maximum activation that activated in fewer than 90% of the sequences. These features represent localized structural or functional motifs rather than global sequence properties. Table 1 summarizes their activation statistics.

Table 1: Activation statistics for the top 5 identified orphan features in ESM-2 layer 4. Frequency is defined as the fraction of positions where activation > 0.1 .

Feature Index	Max Activation	Frequency (> 0.1)
4383	2.32	0.01
9934	1.33	0.01
608	1.30	0.01
716	1.25	0.01
989	1.09	0.02

The most prominent feature, Index 4383, exhibited a peak activation of 2.32, which is significantly higher than the average activation across the SAE. The low frequencies (1–2%) confirm that these features are highly specific to particular sequence contexts.

4.2 LLM-Generated Biological Hypotheses

We utilized GPT-4O to interpret the sequence windows associated with the peak activations of these features. Table 2 presents the hypothesized motifs and the corresponding biophysical rationale.

Table 2: LLM-generated hypotheses for selected SAE features. Sequence windows show 10 residues flanking the peak activation site (indicated in brackets).

Feature	Peak Sequence Window	Hypothesized Motif	Biophysical Rationale
4383	EKGKQSGIIG[A]SLDGTNPAL	Charged Turn Motif	Presence of flanking charged residues (E, K, Q) surrounding a central alanine suggests a propensity for a flexible turn stabilized by electrostatic interactions.
9934	DANAEIEILR[S]RMVVGKAVD	Positively-Charged Helix Cap	The central arginine (R) and surrounding polar residues often act as helix stabilizers, specifically at the capping positions where charge neutralization occurs.
716	EKGKQSGIIG[A]SLDGTNPAL	Charged Beta-Turn	Centered glycine (G) is a common beta-turn initiator; surrounding charged residues provide local stabilization in non-helical regions.

Interestingly, Features 4383 and 716 both peaked on the same sequence segment from the dataset. While Feature 4383 was interpreted as a general "Charged Turn Motif," Feature 716 was characterized more specifically as a "Charged Beta-Turn" due to the emphasis on the glycine residue. This phenomenon, known as "feature splitting" Simon and Zou [2024], suggests that the SAE has learned multiple, slightly different facets of the same underlying structural concept, providing a more granular view of the motif than a single feature would allow.

4.3 Random Baseline Comparison

To verify the validity of the LLM’s interpretations, we performed a baseline test using a random sequence fragment: NPTGLKASLS[P]QRELRAQG. When prompted with this sequence, the LLM hypothesized a "Proline-Induced Structural Disruption." While biophysically plausible (as prolines

often break helices), the hypothesis was more generic and lacked the specific alignment with local charge distributions seen in the SAE-selected features. This indicates that while the LLM is capable of rationalizing any sequence, the SAE performs the critical role of selecting sequences that actually represent conserved, repeating motifs across the model’s internal representation.

5 Discussion

The results of our study demonstrate that protein language models encode biologically coherent information that extends beyond existing human-curated annotations. By isolating ”orphan” SAE features, we have shown that it is possible to identify consistent biophysical patterns that the model utilizes to represent protein sequences.

Interpretation of Orphan Features The identification of motifs like ”Charged Turn Motifs” and ”Positively-Charged Helix Caps” suggests that ESM-2 has internalized fundamental biophysical principles to optimize its next-amino-acid prediction objective. These motifs are essential for protein stability and folding but are often too generic to be formally annotated as functional ”domains” in databases like Pfam. However, for the model, these motifs serve as critical structural landmarks. Our pipeline successfully translates these neural landmarks into human-readable scientific hypotheses.

Limitations Despite the promise of this approach, several limitations remain. **Small Sample Size:** Our current study was conducted on a subset of 100 SWISS-PROT sequences. While this is sufficient for a proof-of-concept, a robust discovery pipeline would require scanning millions of sequences to confirm that these features represent universal biophysical rules rather than artifacts of a small dataset. **LLM Hallucination Risk:** As demonstrated by our random baseline comparison, LLMs like GPT-4O are highly capable of generating plausible-sounding rationales for any sequence fragment. The ”discovery” is therefore heavily dependent on the quality of the SAE’s feature selection. If the SAE identifies a spurious or polysemantic feature, the LLM will still attempt to explain it, potentially leading to false positives. **Lack of Experimental Validation:** Currently, our hypotheses remain purely computational. A true discovery requires laboratory verification, such as site-directed mutagenesis of the identified motifs to observe effects on protein stability or function.

Broader Implications This work contributes to the growing field of ”generative biology,” where AI is used not just to solve known problems (like structure prediction) but to suggest new directions for research. By treating foundation models as active engines for hypothesis generation, we can potentially accelerate the discovery of novel biological mechanisms. Furthermore, the ability to interpret the internal representations of these models is a crucial step toward building ”trustworthy” AI for biology, where we understand the biophysical basis of the model’s predictions.

6 Conclusion

In this paper, we presented a pipeline for the discovery of novel biological mechanisms through the lens of mechanistic interpretability. By combining the disentanglement capabilities of Sparse Autoencoders with the semantic reasoning of Large Language Models, we identified and characterized several ”orphan” features in the ESM-2 protein language model. Our findings suggest that these models learn highly specific biophysical abstractions, such as charged turn motifs and helix stabilizers, which are not currently captured in standard biological annotations.

The ability to bridge the gap between high-dimensional neural activations and human-readable biological hypotheses represents a significant step toward using AI as an active partner in scientific discovery. While our study is a proof-of-concept, it highlights the potential for foundation models to reveal deeper principles of biological organization.

Future work will focus on scaling this pipeline to the entire SWISS-PROT database and implementing automated filters to exclude features that correlate with known experimental structures (e.g., from the PDB). Most importantly, we aim to validate the generated hypotheses through laboratory experiments, testing whether the identified ”orphan” features indeed correspond to critical determinants of protein function and stability.

References

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